Aprendizagem Supervisionada

Thiago Medeiros Barros

Tecnologia em Sistemas para Internet

Agenda

Definição

Exemplos de algoritmos de aprendizagem supervisionada

Pipeline de dados

Divisão do conjunto de Teste, Treino e Validação

Bias/Variance

Avaliação de Modelo

Interpretabilidade

Automl e hiperparâmetros

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Aprendizagem supervisionada

Utiliza o histórico de dados de um fenômeno observado composto por um conjunto de variáveis de entrada (variáveis independentes) e uma variável de saída (variável dependente) associado a eles. **O objetivo dos algoritmos dessa classe é produzir um modelo, hipótese, ou função capaz de relacionar as variáveis de entrada com a variável de saída.**

Exemplo: podemos ter como variáveis de entrada informações sobre alunos (como caracterização social, demográficos e educacionais) e como valor de saída o seu desempenho na disciplina de português. Ao entrar um novo grupo de alunos na instituição de educação, podemos tentar prever o desempenho deles na referida disciplina de acordo com o modelo gerado (treinado) previamente com os dados disponíveis anteriormente.

Quando a variável de saída é do tipo numérico, temos um processo de regressão e, caso a variável de saída seja do tipo categórico, temos um processo de classificação de dados.

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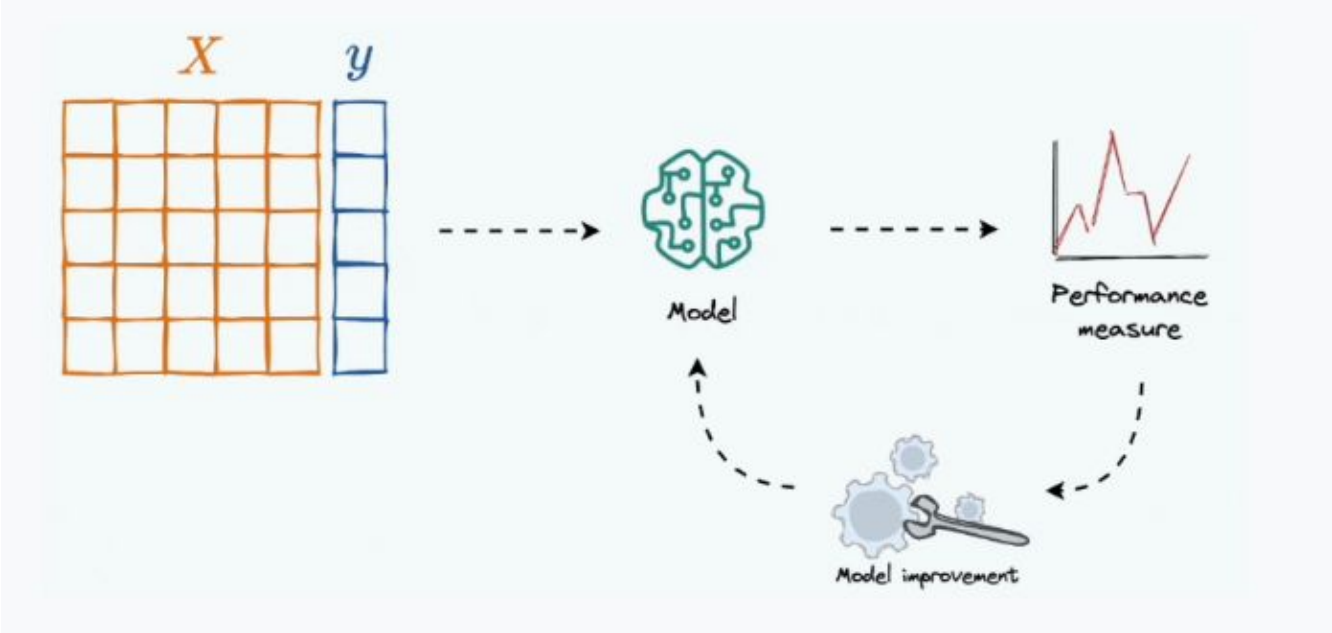

	LnguaPortuguesaeLiteraturaI90H	MatemticaI120H	descricao_responsavel_escolaridade	idade	pessoa_fisica__sexo	classe
0	68.0	66.0	Ensino fundamental incompleto	15	F	1
1	73.0	36.0	Ensino fundamental incompleto	17	M	0
2	61.0	60.0	Ensino médio completo	15	M	1
3	NaN	NaN	Ensino fundamental incompleto	16	F	1
4	69.0	63.0	Ensino fundamental incompleto	15	F	1
...	...	...	...	...	...	...
8903	29.0	11.0	Ensino fundamental incompleto	15	M	0
8904	86.0	71.0	Ensino fundamental incompleto	15	F	1
8905	61.0	67.0	Ensino fundamental incompleto	15	F	1
8906	93.0	78.0	Ensino fundamental incompleto	14	F	1
8907	81.0	63.0	Ensino fundamental incompleto	15	M	1

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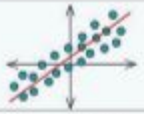
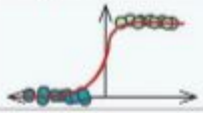
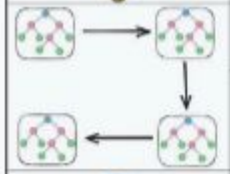
Loss functions of 16 Most Popular ML Algorithms

blog.DailyDoseofDS.com

	Linear Regression	Mean Squared Error
	Logistic Regression	Cross-Entropy Loss
	Decision Tree Classifier	Information Gain or Gini impurity
	Decision Tree Regressor	Mean Squared Error
	Random Forest Classifier	Information Gain or Gini impurity
	Random Forest Regressor	Mean Squared Error
	Support Vector Machines (SVMs)	Hinge Loss
	k-Nearest Neighbors	No loss function
$P(B\|A) = \frac{P(B \cap A)}{P(A)}$	Naive Bayes	No loss function
	Neural Networks	Regression: Mean Squared Error Classification: Cross-Entropy Loss
	AdaBoost	Exponential loss
	Gradient Boosting | LightGBM | CatBoost | XGBoost	Regression: Mean Squared Error Classification: Cross-Entropy Loss
	KMeans Clustering	??

*These are typically used loss functions. It does not mean they are used always in these algorithms.

10 Most Common Loss Functions in Machine Learning

blog.DailyDoseofDS.com

Loss Function Name	Description	Function
Regression Loss Functions		
Mean Bias Error	Captures average bias in prediction. But is rarely used for training.	$\mathcal{L}_{MBE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - f(x_i))$
Mean Absolute Error	Measures absolute average bias in prediction. Also called L1 Loss.	$\mathcal{L}_{MAE} = \frac{1}{N}\sum_{i=1}^{N}\|y_i - f(x_i)\|$
Mean Squared Error	Average squared distance between actual and predicted. Also called L2 Loss.	$\mathcal{L}_{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - f(x_i))^2$
Root Mean Squared Error	Square root of MSE. Loss and dependent variable have same units.	$\mathcal{L}_{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - f(x_i))^2}$
Huber Loss	A combination of MSE and MAE. It is parametric loss function.	$\mathcal{L}_{\text{Huberloss}} = \begin{cases} \frac{1}{2}(y_i - f(x_i))^2 & : \|y_i - f(x_i)\| \le \delta \\ \delta(\|y_i - f(x_i)\| - \frac{1}{2}\delta) & : otherwise \end{cases}$
Log Cosh Loss	Similar to Huber Loss + non-parametric. But computationally expensive.	$\mathcal{L}_{\text{LogCosh}} = \frac{1}{N}\sum_{i=1}^{N} log(cosh(f(x_i) - y_i))$
Classification Loss Functions (Binary + Multi-class)		
Binary Cross Entropy (BCE)	Loss function for binary classification tasks.	$\mathcal{L}_{BCE} = \frac{1}{N}\sum_{i=1}^{N} y_i \cdot log(p(x_i)) + (1 - y_i) \cdot log(1 - p(x_i))$
Hinge Loss	Penalizes wrong and right (but less confident) predictions. Commonly used in SVMs.	$\mathcal{L}_{\text{Hinge}} = max(0, 1 - (f(x) \cdot y))$
Cross Entropy Loss	Extension of BCE loss to multi-class classification.	$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M} y_{ij} log(f(x_{ij}))$ $N : samples; M : classes$
KL Divergence	Minimizes the divergence between predicted and true probability distribution	$\mathcal{L}_{KL} = \sum_{i=1}^{N} y_i \cdot log(\frac{y_i}{f(x_i)})$

https://blog.dailydoseofds.com/p/free-daily-dose-of-data-science-pdf

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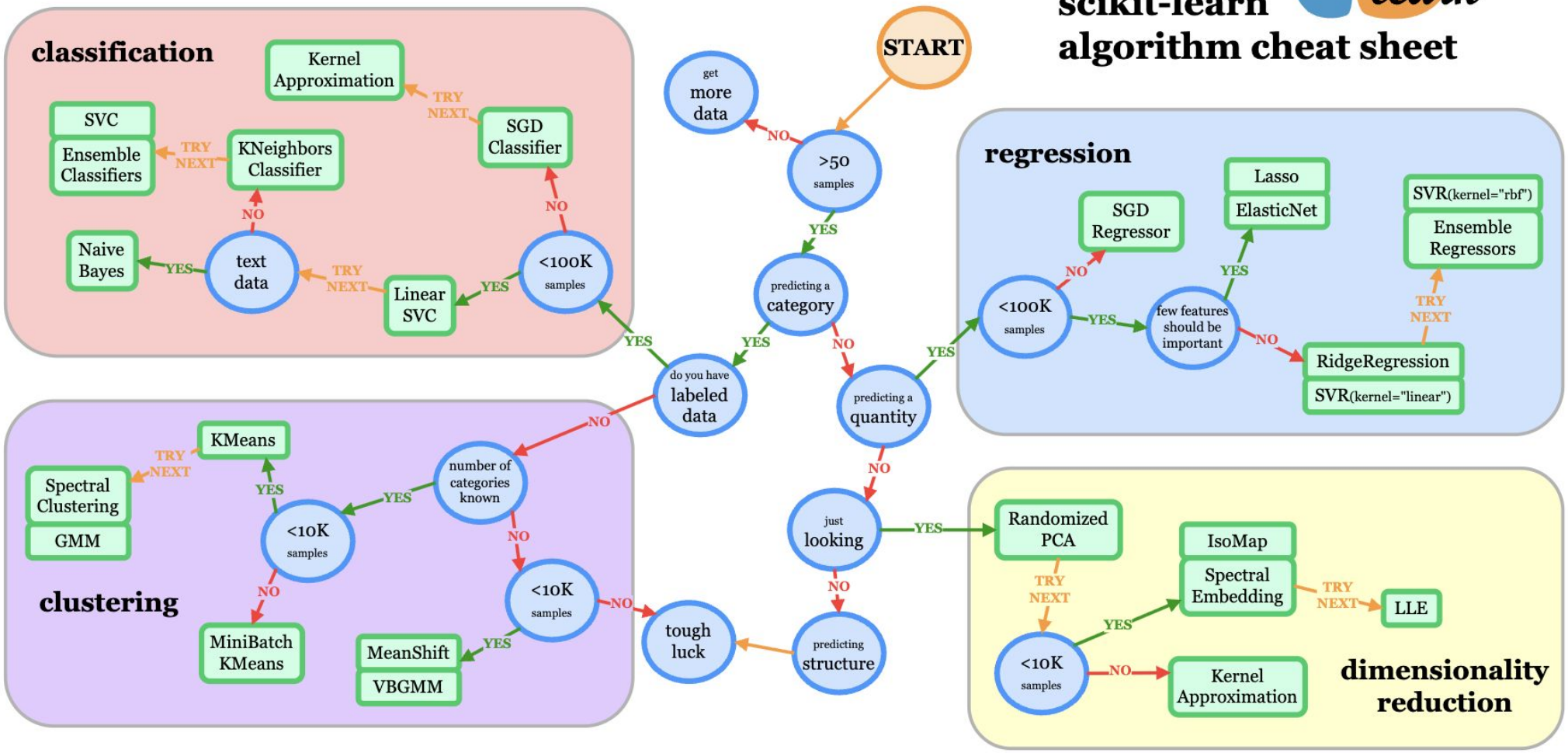
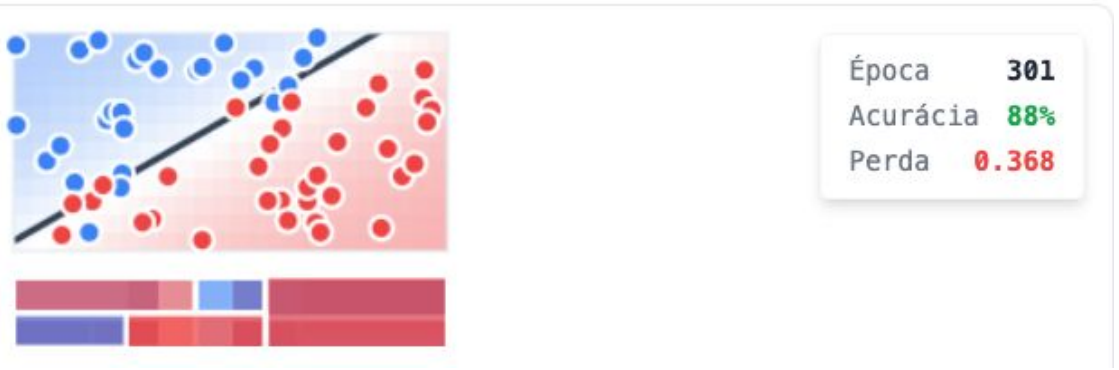
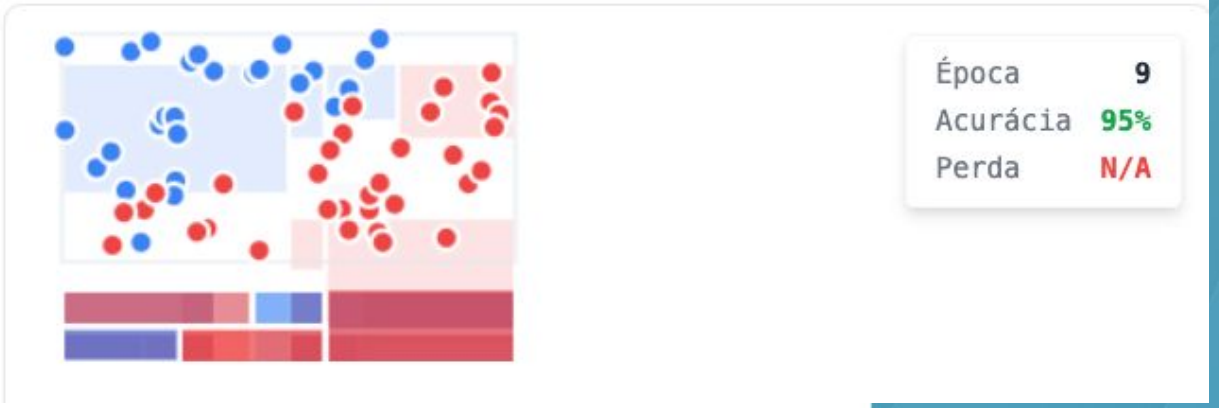
scikit-learn algorithm cheat sheet



Regressão Logística vs Árvore de Decisão

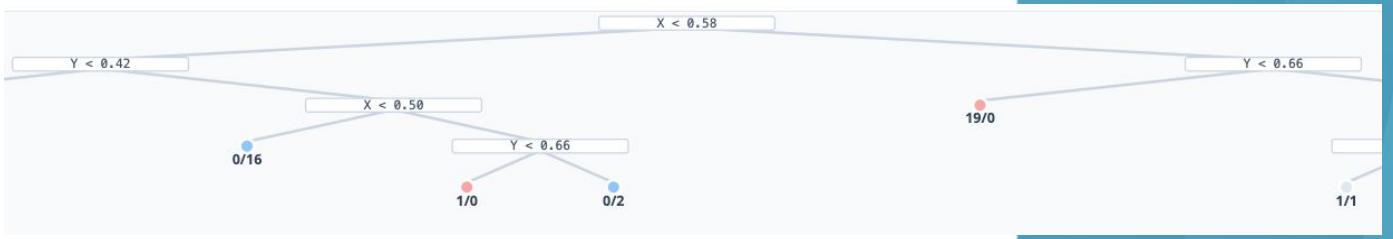




PESO X PESO Y BIAS

$$z = -3.21 \cdot x + +2.80 \cdot y + -0.13$$

Probabilidade (Azul) = $1 / (1 + e^{-z})$



https://gemini.google.com/share/b28dcad346b8

Teachable Machine

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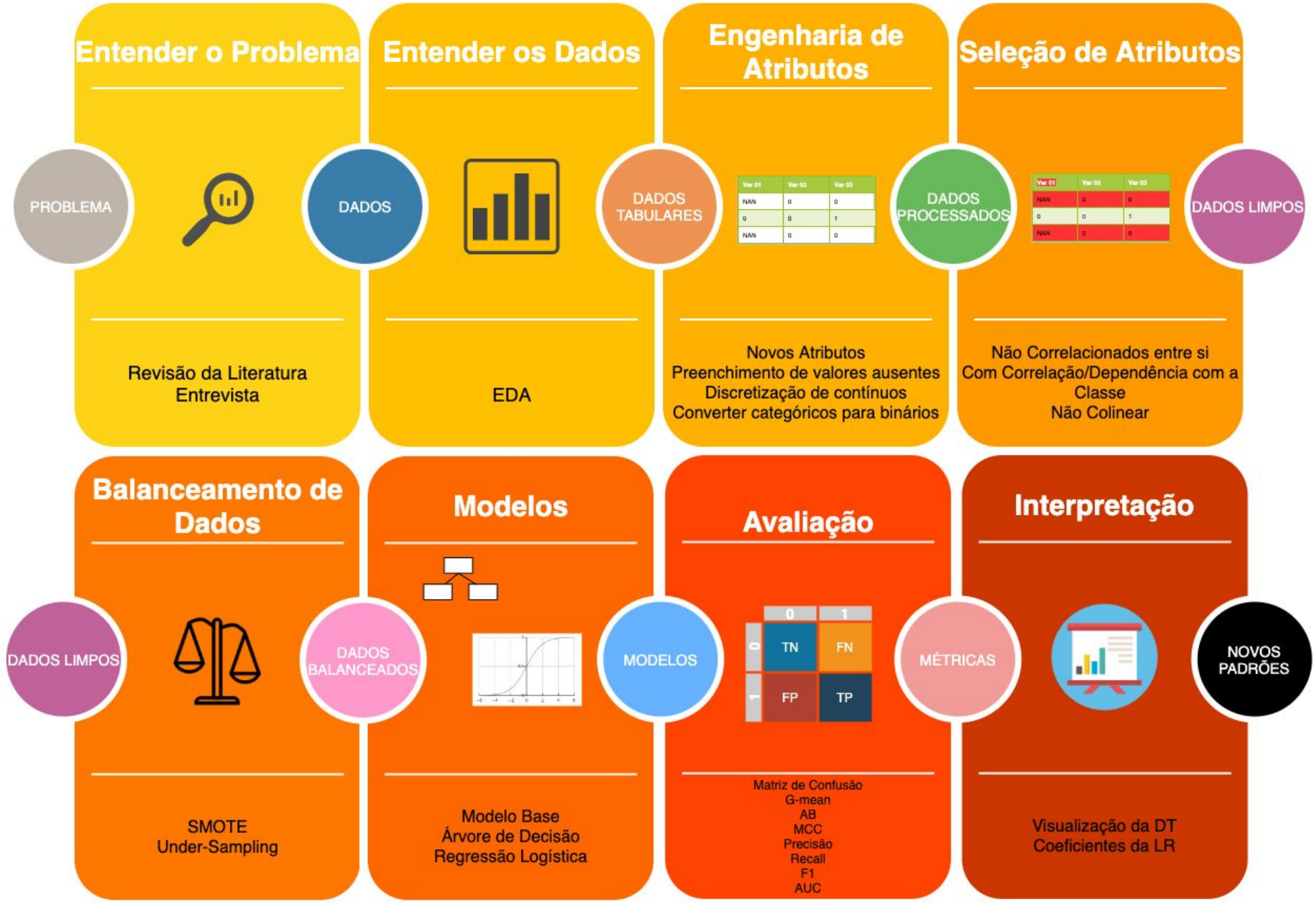
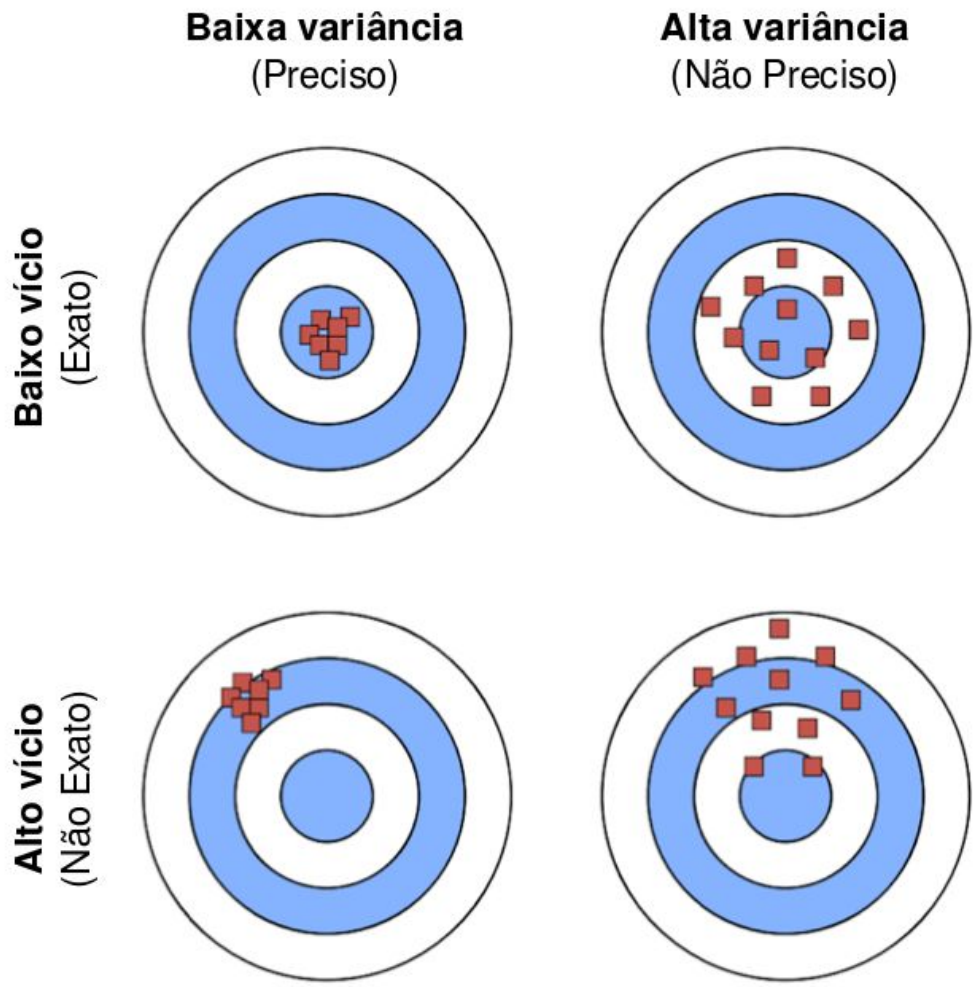
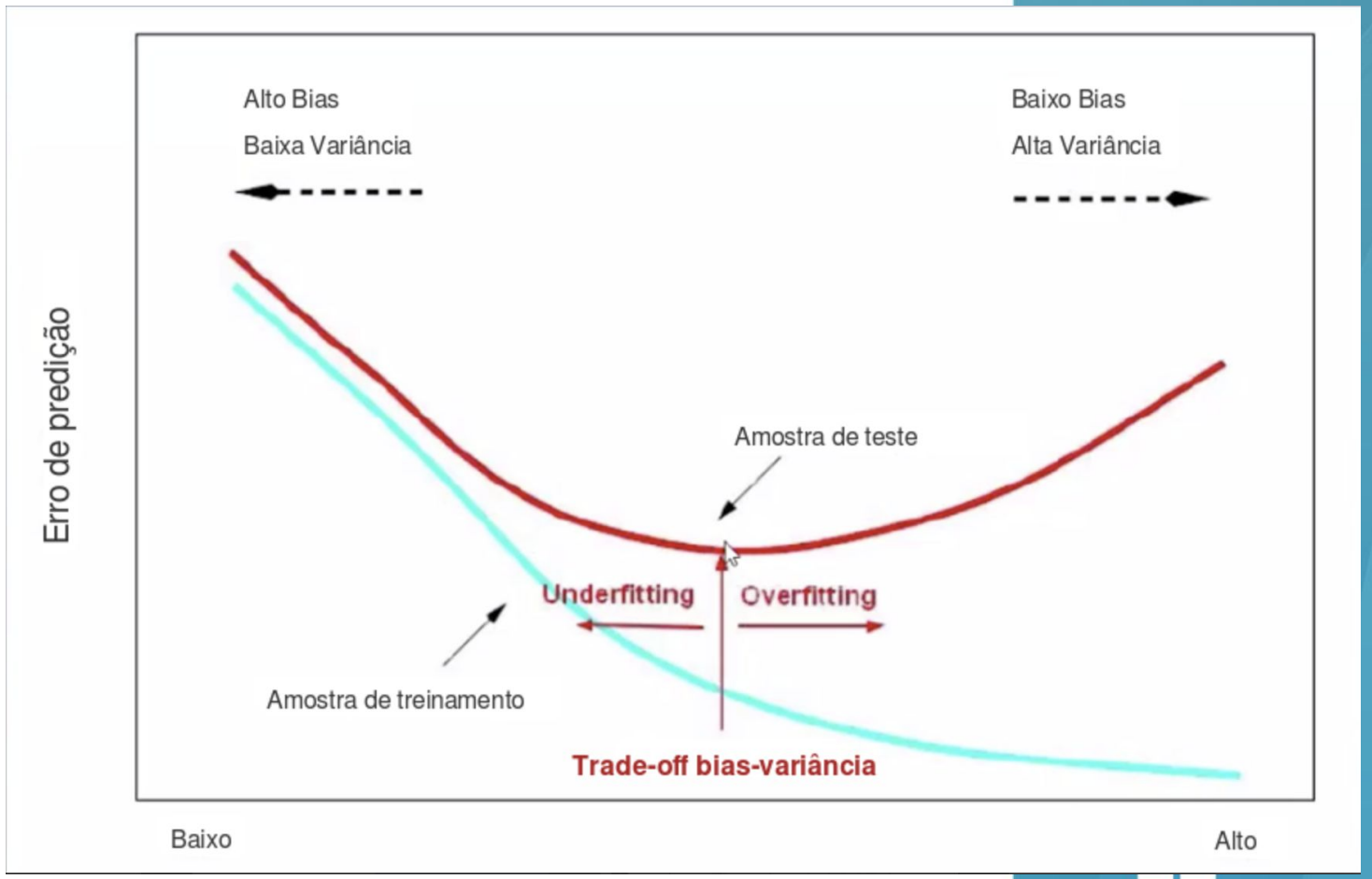
https://teachablemachine.withgoogle.com/


Entender o Problema
PROBLEMA
DADOS
Revisão da Literatura
Entrevista
Entender os Dados
DADOS TABULARES
EDA
Engenharia de Atributos
DADOS PROCESSADOS
Novos Atributos
Preenchimento de valores ausentes
Discretização de contínuos
Converter categóricos para binários
Seleção de Atributos
DADOS LIMPOS
Não Correlacionados entre si
Com Correlação/Dependência com a Classe
Não Colinear
Balanceamento de Dados
DADOS LIMPOS
DADOS BALANCEADOS
SMOTE
Under-Sampling
Modelos
MODELOS
Modelo Base
Árvore de Decisão
Regressão Logística
Avaliação
MÉTRICAS
0
1
TN
FN
FP
TP
Matriz de Confusão
G-mean
AB
MCC
Precisão
Recall
F1
AUC
Interpretação
NOVOS PADRÕES
Visualização da DT
Coeficientes da LR

Split

https://github.com/ivanovitchm/ppgeecmachinelearning


Baixa variância
(Preciso)
Alta variância
(Não Preciso)
Baixo vício
(Exato)
Alto vício
(Não Exato)


Alto Bias
Baixa Variância
Baixo Bias
Alta Variância
Erro de predição
Amostra de teste
Underfitting
Overfitting
Amostra de treinamento
Trade-off bias-variância
Baixo
Alto

Matriz de Confusão

Predicted Label (Predito) \ True Label (Classe Real)	1 (Positivo)	0 (Negativo)
1	TP: 410 (Verdadeiro Positivo)	FP: 100 (Falso Positivo)
0	FN: 90 (Falso Negativo)	TN: 400 (Verdadeiro Negativo)

TOTAL DE AMOSTRAS	ACURÁCIA GLOBAL
1000	81.0%

Precisão (Precision) — PPV

$$\frac{TP}{TP + FP}$$

$$\frac{410}{410 + 100}$$

0.8039

Recall (Sensibilidade) — TPR

$$\frac{TP}{TP + FN}$$

$$\frac{410}{410 + 90}$$

0.8200

F1 Score — Harmonic

$$2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$2 * \frac{0.80 * 0.82}{0.80 + 0.82}$$

0.8119

G-mean — Geometric

$$\sqrt{\text{Recall} * \text{Specificity}}$$

$$\sqrt{0.820 * 0.800}$$

0.8099

Matriz de Confusão

Predicted Label (Predito) \ True Label (Classe Real)	1 (Positivo)	0 (Negativo)
1	TP: 10 (Verdadeiro Positivo)	FP: 1 (Falso Positivo)
0	FN: 90 (Falso Negativo)	TN: 899 (Verdadeiro Negativo)

TOTAL DE AMOSTRAS	ACURÁCIA GLOBAL
1000	90.9%

Precisão (Precision) — PPV

$$\frac{TP}{TP + FP}$$

$$\frac{10}{10 + 1}$$

0.9091

Recall (Sensibilidade) — TPR

$$\frac{TP}{TP + FN}$$

$$\frac{10}{10 + 90}$$

0.1000

F1 Score — Harmonic

$$2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$2 * \frac{0.91 * 0.10}{0.91 + 0.10}$$

0.1802

G-mean — Geometric

$$\sqrt{\text{Recall} * \text{Specificity}}$$

$$\sqrt{0.100 * 0.999}$$

0.3161

https://gemini.google.com/share/fa839209d906

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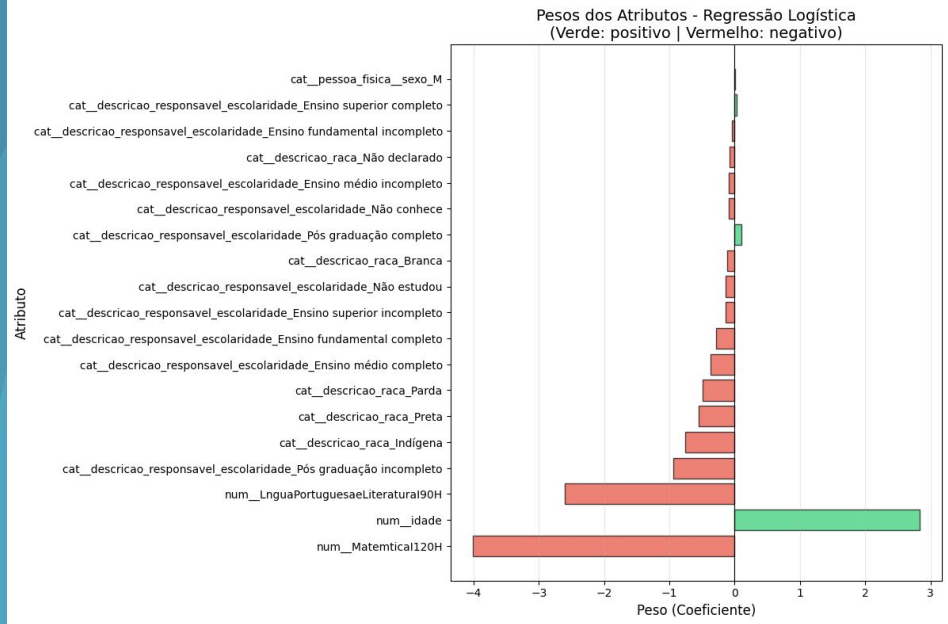
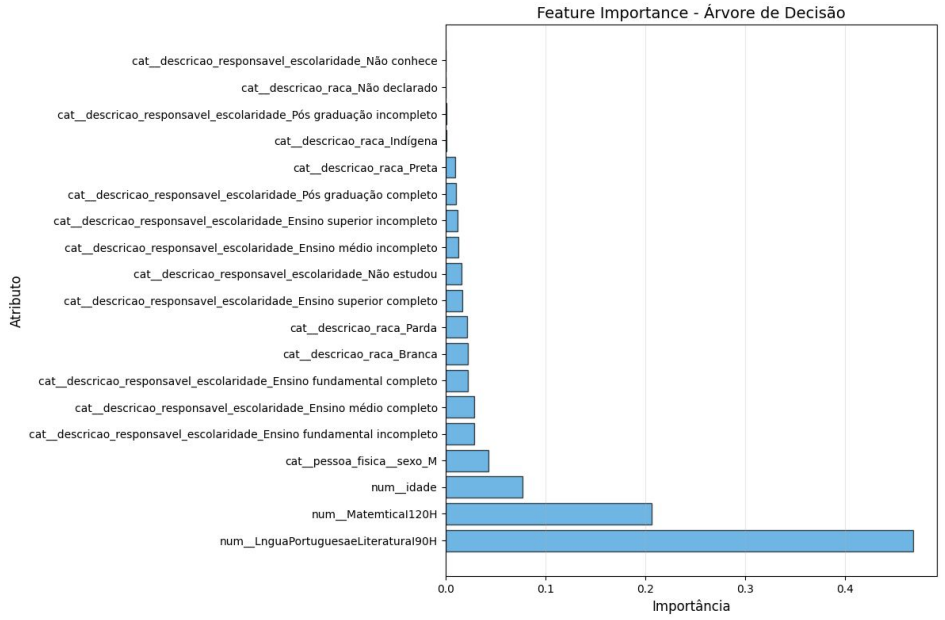
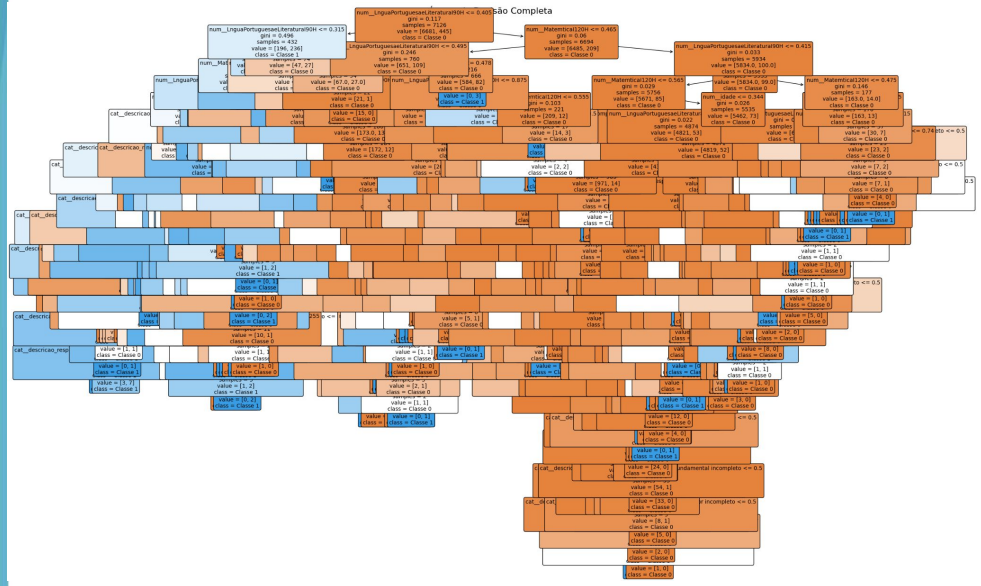
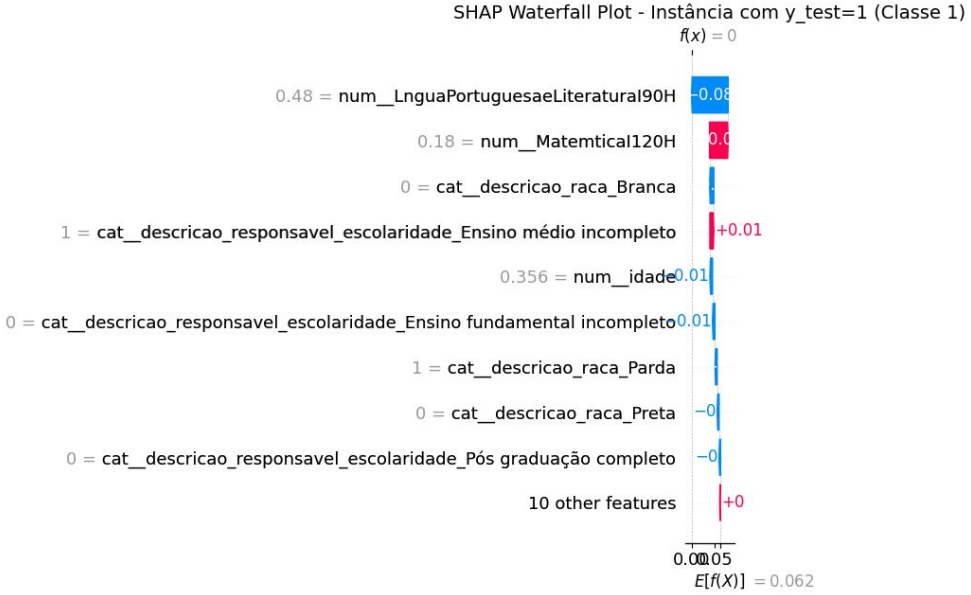
Interpretabilidade



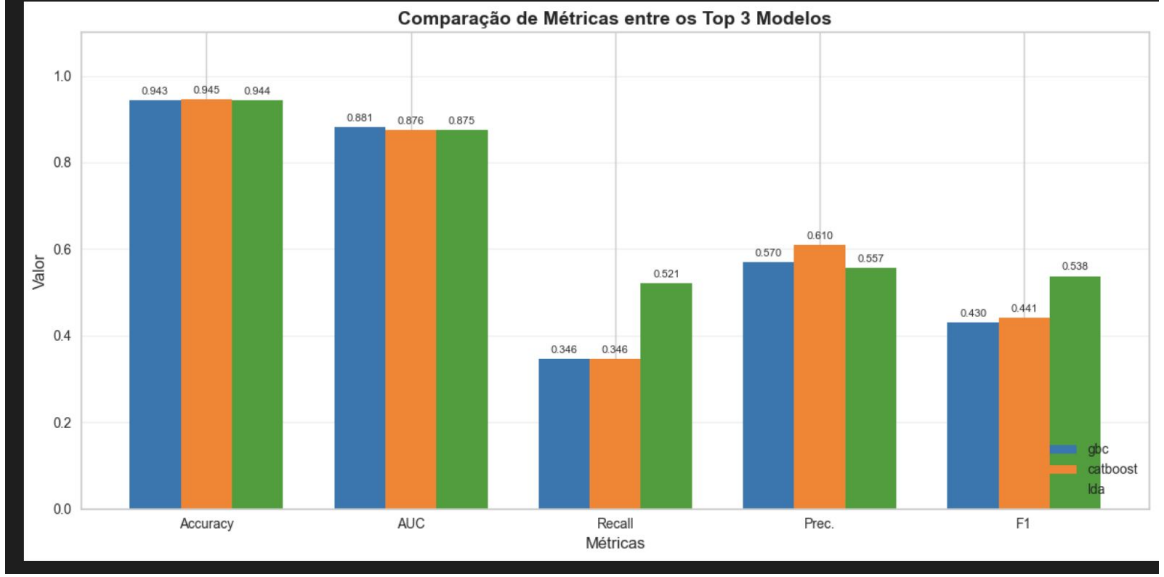






Automl e hiperparâmetros

	Model	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC	TT (Sec)
gbc	Gradient Boosting Classifier	0.9427	0.8815	0.3461	0.5704	0.4301	0.4019	0.4162	0.058
catboost	CatBoost Classifier	0.9453	0.8758	0.3461	0.6103	0.4408	0.4144	0.4333	0.366
lda	Linear Discriminant Analysis	0.9441	0.8750	0.5213	0.5567	0.5377	0.5080	0.5087	0.010
ridge	Ridge Classifier	0.9399	0.8742	0.1663	0.5586	0.2532	0.2321	0.2802	0.012
ada	Ada Boost Classifier	0.9413	0.8739	0.3416	0.5504	0.4205	0.3914	0.4043	0.242
lr	Logistic Regression	0.9432	0.8724	0.3056	0.5874	0.3994	0.3729	0.3961	0.442
lightgbm	Light Gradient Boosting Machine	0.9441	0.8623	0.3798	0.5849	0.4594	0.4313	0.4431	0.464
rf	Random Forest Classifier	0.9446	0.8544	0.3371	0.6027	0.4319	0.4052	0.4244	0.254
svm	SVM - Linear Kernel	0.9413	0.8512	0.3101	0.5512	0.3839	0.3574	0.3782	0.010
xgboost	Extreme Gradient Boosting	0.9422	0.8510	0.3596	0.5605	0.4370	0.4081	0.4198	0.018
nb	Naive Bayes	0.7245	0.8395	0.6697	0.2554	0.3441	0.2772	0.2894	0.258
et	Extra Trees Classifier	0.9415	0.8219	0.3303	0.5516	0.4114	0.3828	0.3976	0.038
knn	K Neighbors Classifier	0.9423	0.7716	0.2831	0.5770	0.3794	0.3529	0.3779	0.258
dt	Decision Tree Classifier	0.9189	0.6774	0.3910	0.3637	0.3756	0.3324	0.3333	0.234
qda	Quadratic Discriminant Analysis	0.6181	0.6533	0.8022	0.1606	0.2576	0.1708	0.2306	0.012
dummy	Dummy Classifier	0.9376	0.5000	0.0000	0.0000	0.0000	0.0000	0.0000	0.010



	Accuracy	AUC	Recall	Prec.	F1	Kappa	MCC
Fold							
0	0.9453	0.8786	0.4270	0.5846	0.4935	0.4653	0.4718
1	0.9418	0.8696	0.3483	0.5536	0.4276	0.3986	0.4105
2	0.9460	0.8817	0.4045	0.6000	0.4832	0.4559	0.4657
3	0.9481	0.9183	0.3933	0.6364	0.4861	0.4604	0.4752
4	0.9432	0.8782	0.4045	0.5625	0.4706	0.4414	0.4481
Mean	0.9448	0.8853	0.3955	0.5874	0.4722	0.4443	0.4543
Std	0.0022	0.0170	0.0260	0.0294	0.0235	0.0242	0.0238

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits

Melhor modelo após tuning: GradientBoostingClassifier(ccp_alpha=0.0, criterion='friedman_mse', init=None,
                           learning_rate=0.05, loss='log_loss', max_depth=6,
                           max_features='sqrt', max_leaf_nodes=None,
                           min_impurity_decrease=0.3, min_samples_leaf=4,
                           min_samples_split=10, min_weight_fraction_leaf=0.0,
                           n_estimators=270, n_iter_no_change=None,
                           random_state=42, subsample=0.7, tol=0.0001,
                           validation_fraction=0.1, verbose=0,
                           warm_start=False)

Hiperparâmetros otimizados:
  ccp_alpha: 0.0
  criterion: friedman_mse
  init: None
  learning_rate: 0.05
  loss: log_loss
  max_depth: 6
  max_features: sqrt
  max_leaf_nodes: None
  min_impurity_decrease: 0.3
  min_samples_leaf: 4
  min_samples_split: 10
  min_weight_fraction_leaf: 0.0
```

Vamos praticar

https://github.com/tmedeirosb/tsi-ad-2025/blob/main/MODULO_08.ipynb

https://colab.research.google.com/github/tmedeirosb/tsi-ad-2025/blob/main/MODULO_08.ipynb

https://tsi-ad-2025-predict.streamlit.app/

https://colab.research.google.com/github/tmedeirosb/tsi-ad-2025/blob/main/MODULO_08_PYCARET.ipynb

https://colab.research.google.com/github/tmedeirosb/tsi-ad-2025/blob/main/MODULO_08_AUTOGLUON.ipynb

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